Analysis_Jan2025_results_writeup

2025-12-09

```r
library(data.table)
library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.2 --
## v ggplot2 4.0.1     v purrr   1.1.0
## v tibble  3.3.0     v dplyr   1.1.4
## v tidyr   1.3.1     v stringr 1.5.1
## v readr   2.1.2     v forcats 0.5.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::between()   masks data.table::between()
## x dplyr::filter()    masks stats::filter()
## x dplyr::first()     masks data.table::first()
## x dplyr::lag()       masks stats::lag()
## x dplyr::last()      masks data.table::last()
## x purrr::transpose() masks data.table::transpose()
```

```r
library(ggplot2)
library(here)
```

```
## here() starts at /Users/nencheva/Library/CloudStorage/GoogleDrive-mira.nencheva@gmail.com/.shortcut-
```

```r
# Reproducible file reading
affect_ratings <- data.table(read_csv(here("Data", "affect_ratings_merged_with_marker_id.csv")))
```

```
## Rows: 1081 Columns: 12
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): file_name, lab_id, subj_id, trial_num
## dbl (8): folder, affect_rating_segment, segment_start_sec, segment_end_sec, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
main_data <- data.table(read_csv(here("Data", "clean_data.csv"))) %>%
  select(lab_id,subj_id,age_days, age_group,condition,cb_order,helper_hinderer_choice,
       push_up_identity, push_up_order, push_up_side, meet_eligibility)
```

```
## Rows: 1018 Columns: 102
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (52): lab_id, subj_id, age_group, method, RA_1, RA_1_naive, RA_2, RA_2_n...
```

```
## dbl (50): subj_num, age_days, premature_days, screen_size_inches, infant_dis...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
merged_data <- data.table(merge(affect_ratings,main_data, by = c('lab_id','subj_id')))
```

```r
trial_level_affect <- merged_data[
  ,
  {
    # total duration of coded window for this trial
    total_dur <- 10#sum(duration_sec, na.rm = TRUE)

    # duration (in sec) for positive, negative and any non-neutral
    non_neutral_dur <- sum(duration_sec[!is.na(affect_rating_segment) &
                                          affect_rating_segment != 4],
                           na.rm = TRUE)
    pos_dur <- sum(duration_sec[!is.na(affect_rating_segment) &
                                  affect_rating_segment > 4],
                   na.rm = TRUE)
    neg_dur <- sum(duration_sec[!is.na(affect_rating_segment) &
                                  affect_rating_segment < 4],
                   na.rm = TRUE)

    # centered affect (neutral = 0)
    affect_centered <- affect_rating_segment - 4

    # duration-weighted affect rating average INCLUDING neutral
    affect_avg <- if (total_dur > 0) {
      sum(affect_centered * duration_sec, na.rm = TRUE) / total_dur
    } else {
      NA_real_
    }

    # duration-weighted average EXCLUDING neutral (0s -> NA)
    affect_avg_nonneutral <- {
      affect_centered_nz <- fifelse(affect_centered == 0, NA_real_, affect_centered)

      nonneutral_dur <- sum(duration_sec[!is.na(affect_centered_nz)], na.rm = TRUE)

      if (nonneutral_dur > 0) {
        sum(affect_centered_nz * duration_sec, na.rm = TRUE) / nonneutral_dur
      } else {
        NA_real_
      }
    }


    # get first reaction
    first_idx_any <- {
      ok <- which(!is.na(segment_start_sec) & !is.na(affect_centered) & !is.na(duration_sec))
      if (length(ok) == 0) NA_integer_ else ok[which.min(segment_start_sec[ok])]
    }
```

```r
# find first affect reaction
first_affect <- if (!is.na(first_idx_any)) affect_centered[first_idx_any] else NA_real_




# Peak positive / negative ratings within trial (exploratory measure)
peak_pos <- {
  v <- affect_centered[!is.na(affect_centered)]
  if (length(v) == 0) NA_real_ else max(v)
}

peak_neg <- {
  v <- affect_centered[!is.na(affect_centered)]
  if (length(v) == 0) NA_real_ else min(v)
}

# Strongest affect regardless of sign
peak_abs <- {
  v <- affect_centered[!is.na(affect_centered)]
  if (length(v) == 0) NA_real_ else max(abs(v))
}

peak_signed <- {
  v <- affect_centered[!is.na(affect_centered)]
  if (length(v) == 0) {
    NA_real_
  } else {
    v[which.max(abs(v))]   # keeps the sign of the strongest deviation
  }
}




# take first values per trial for video-level categorical ratings
happiness_val <- first(happiness_full_video)
surprise_val  <- first(surprise_full_video)

.(
  # keep other trial-level columns (take first value within trial)
  folder                  = first(folder),
  file_name               = first(file_name),
  condition               = first(condition),
  cb_order                = first(cb_order),
  helper_hinderer_choice  = first(helper_hinderer_choice),
  push_up_identity        = first(push_up_identity),
  push_up_order           = first(push_up_order),
  push_up_side            = first(push_up_side),
  meet_eligibility        = first(meet_eligibility),

  #affectPresent_duration: this is the cumulative duration of non-neutral expressions in seconds ov
  affectPresent_duration = if (total_dur > 0) non_neutral_dur / total_dur else NA_real_,
```

```r
      #affectPositive_duration: this is the cumulative duration of positive expressions in seconds over
      affectPositive_duration = if (total_dur > 0) pos_dur / total_dur else NA_real_,

      #affectNegative_duration: this is the cumulative duration of negative expressions in seconds over
      affectNegative_duration = if (total_dur > 0) neg_dur / total_dur else NA_real_,

      #affect_average: this is the weighted average of the Likert ratings over the analysis window, thi
      affect_average = affect_avg,
      affect_average_nonneutral = affect_avg_nonneutral,
      # affect rating of first reaction for the trial (exploratory measure)
      first_affect = first_affect,
      # peak affect rating for the trial (exploratory measure)
      peak_affect_rating_abs = peak_abs,
      peak_affect_rating_signed = peak_signed,
      #affectPositive_categoricalChoice: this is the categorical choice affect rating, coded as 1 for p
      affectPositive_categoricalChoice = if (is.na(happiness_val)) NA_integer_ else if (happiness_val ==
      #affectNegative_categoricalChoice: this is the categorical choice affect rating, coded as 1 for n
      affectNegative_categoricalChoice = if (is.na(happiness_val)) NA_integer_ else if (happiness_val ==
      #surprise_categoricalChoice: this is the categorical choice surprise rating, coded as 1 for surpr
      surprise_categoricalChoice = if (is.na(surprise_val)) NA_integer_ else if (surprise_val == 1) 1L
    )
  },
  by = .(lab_id, subj_id, trial_num)
]



# tag push up and push down trials
trial_level_affect[, trial_num_int := as.integer(trial_num)]
trial_level_affect[
  ,
  is_push_up :=
    (push_up_order == "first"  & trial_num_int %% 2L == 1L) |
    (push_up_order == "second" & trial_num_int %% 2L == 0L)
]

trial_level_affect[
  ,
  push_direction := ifelse(is_push_up, "push_up", "push_down")
]



# Rename variables to align them with preregistered models
trial_level_affect[, social := factor(condition)]
trial_level_affect[
  ,
  helping := factor(
    ifelse(is.na(is_push_up), NA_character_,
           ifelse(is_push_up, "help", "hinder")),
    levels = c("hinder", "help")
  )
]
```

```r
trial_level_affect[
  ,
  chose_helper := fifelse(
    helper_hinderer_choice == "helper",   1L,
    fifelse(helper_hinderer_choice == "hinderer", 0L, NA_integer_)
  )
]
```

```r
library(lmerTest)
```

```
## Loading required package: lme4

## Loading required package: Matrix

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

##
## Attaching package: 'lmerTest'

## The following object is masked from 'package:lme4':
##
##     lmer

## The following object is masked from 'package:stats':
##
##     step
```

```r
library(papaja)
```

```
## Loading required package: tinylabels
```

```r
# affectPresent_duration ~ social + (1 | participant_id) + (1 | lab_id)
m_affectPresent_social <- lmer(
  affectPresent_duration ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect
)





m_affectPresent_social_first2trials <- lmer(
  affectPresent_duration ~ social + (1 | subj_id) + (1 | lab_id),
```

```r
  data = trial_level_affect[trial_num_int <= 2 ,]
)


p_affectPresent_social <- ggplot(
  trial_level_affect,
  aes(x = social, y = affectPresent_duration)
) +
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Affect present (duration proportion of trial)"
  )


# affect_average ~ social + (1 | participant_id) + (1 | lab_id)
m_affectAverage_social <- lmer(
  affect_average ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect
)


m_affectAverage_social_first2trials <- lmer(
  affect_average ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect[trial_num_int <= 2 ,]
)



#excluding neutral affect
m_affectAverage_social_nonneutral <- lmer(
  affect_average_nonneutral ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect
)



p_affectAverage_social <- ggplot(
  trial_level_affect,
  aes(x = social, y = affect_average)
) +
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
```

```
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Average affect rating"
  )

apa_m_affectPresent_social <- apa_print(m_affectPresent_social)
apa_m_affectAverage_social <- apa_print(m_affectAverage_social)
apa_m_affectAverage_social_nonneutral <- apa_print(m_affectAverage_social_nonneutral)

#print(paste("affectPresent_duration ~ social:", apa_m_affectPresent_social$full_result$socialsocial))

#print(paste("affect_average ~ social:", apa_m_affectAverage_social$full_result$socialsocial))
```

Social versus non-social condition

First, we tested whether emotional expressions differed between the social and the non-social condition. There was no significant difference in the overall duration of emotion displays (positive or negative) between the social vs nonsocial condition (`affectPresent_duration ~ social + (1 | subj_id) + (1 | lab_id)`: $\hat{\beta} = 0.02$, 95% CI $[-0.04, 0.08]$, $t(92.17) = 0.57$, $p = .573$). Similarly, we did not find evidence that emotional expressions were more positive in the social than the non-social condition (`affect_average ~ social + (1 | subj_id) + (1 | lab_id)`: $\hat{\beta} = 0.05$, 95% CI $[-0.04, 0.14]$, $t(95.53) = 1.14$, $p = .257$), or that they were less neutral (`affect_average_nonneutral ~ social + (1 | subj_id) + (1 | lab_id)`:$\hat{\beta} = 0.19$, 95% CI $[-0.23, 0.61]$, $t(74.41) = 0.87$, $p = .385$). Together these results suggest that infants displayed similar emotions in the social and non-social conditions.

```
library(patchwork)
p_affectPresent_social + p_affectAverage_social
```

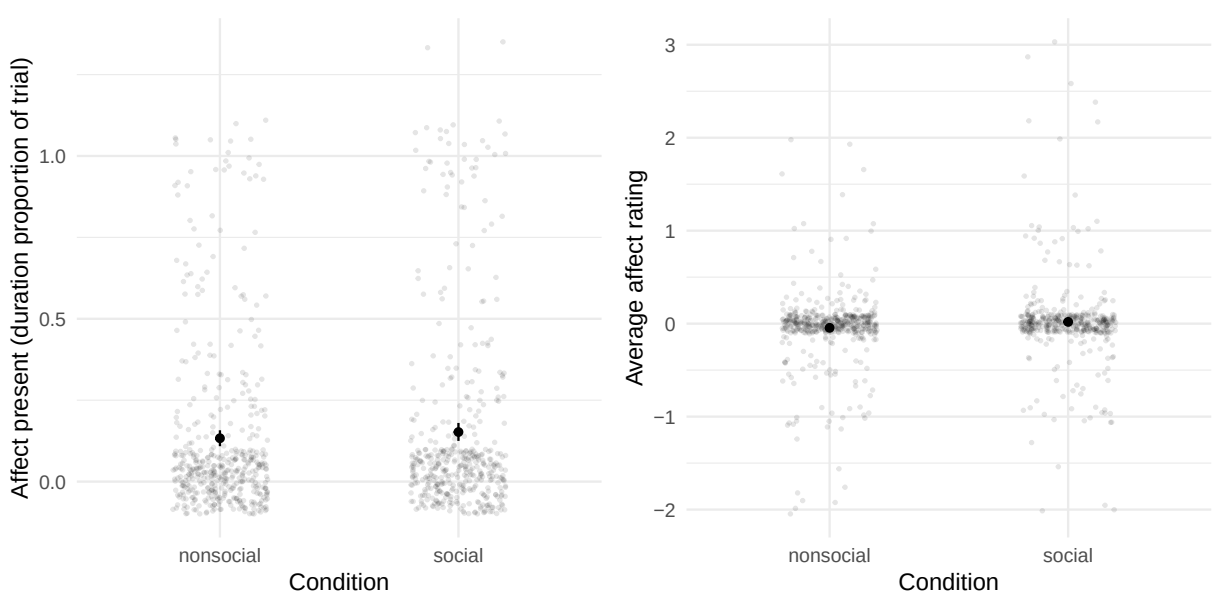


Figure 1: Emotion duration (left) and average affect (right) by social condition

```r
# affect_average ~ helping*social + (helping | participant_id) + (helping | lab_id)
m_affectAverage_helping_social <- lmer(
  affect_average ~ helping * social +
    (helping | subj_id) +
    (1 | lab_id),
  data = trial_level_affect
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
m_affectAverage_helping_social_first2 <- lmer(
  affect_average ~ helping * social +
    (1 | subj_id) +
    (1 | lab_id),
  data = trial_level_affect[trial_num_int <= 2,]
)


apa_m_affectAverage_helping_social <- apa_print(m_affectAverage_helping_social)

print("affect_average ~ helping*social:")
```

```
## [1] "affect_average ~ helping*social:"
```

```r
apa_m_affectAverage_helping_social$table
```

```
## A data.frame with 6 labelled columns:
##
##                                 term estimate      conf.int statistic     df
## 1                          Intercept    -0.08 [-0.17, 0.01]     -1.71   7.47
## 2                        Helpinghelp     0.07  [0.00, 0.15]      1.96 785.65
## 3                       Socialsocial     0.10 [-0.01, 0.20]      1.81 107.51
## 4 Helpinghelp $\\times$ Socialsocial    -0.09 [-0.19, 0.02]     -1.60 785.81
##   p.value
## 1    .128
## 2    .051
## 3    .072
## 4    .111
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $t$
## df       : $\\mathit{df}$
## p.value  : $p$
```

```r
p_affectAverage_helping <- ggplot(
  trial_level_affect,
  aes(x = helping, y = affect_average)
) +
  facet_grid(cols = vars(social))+
```

```r
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Average affect rating"
  )



# affectPositive_duration ~ helping*social + (helping | participant_id) + (helping | lab_id)
m_affectPositive_helping_social <- lmer(
  affectPositive_duration ~ helping * social +
    (1 | subj_id),
  data = trial_level_affect
)

apa_m_affectPositive_helping_social <- apa_print(m_affectPositive_helping_social)

print("affectPositive_duration ~ helping*social:")
```

```
## [1] "affectPositive_duration ~ helping*social:"
```

```r
apa_m_affectPositive_helping_social$table
```

```
## A data.frame with 6 labelled columns:
##
##                                  term estimate      conf.int statistic     df
## 1                           Intercept     0.04  [0.01, 0.07]      2.58 160.76
## 2                         Helpinghelp     0.01 [-0.02, 0.04]      0.73 814.36
## 3                        Socialsocial     0.02 [-0.02, 0.07]      0.96 165.92
## 4 Helpinghelp $\\times$ Socialsocial     0.00 [-0.05, 0.04]     -0.10 814.52
##   p.value
## 1    .011
## 2    .465
## 3    .337
## 4    .923
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $t$
## df       : $\\mathit{df}$
## p.value  : $p$
```

```r
p_affectPositive_helping <- ggplot(
  trial_level_affect,
  aes(x = helping, y = affectPositive_duration)
```

```r
) +
  facet_grid(cols = vars(social))+
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Positive affect duration"
  )




# affectNegative_duration ~ helping*social + (helping | participant_id) + (helping | lab_id)
m_affectNegative_helping_social <- lmer(
  affectNegative_duration ~ helping * social +
    (1 | subj_id) +
    (1 | lab_id),
  data = trial_level_affect
)







apa_m_affectNegative_helping_social <- apa_print(m_affectNegative_helping_social)

print("affectNegative_duration ~ helping*social:")
```

```
## [1] "affectNegative_duration ~ helping*social:"
```

```r
apa_m_affectNegative_helping_social$table
```

```
## A data.frame with 6 labelled columns:
##
##                                 term estimate       conf.int statistic     df
## 1                          Intercept     0.11   [0.05, 0.17]      3.53   5.20
## 2                        Helpinghelp    -0.04 [-0.08, -0.01]     -2.26 811.43
## 3                       Socialsocial    -0.04  [-0.09, 0.02]     -1.38 166.04
## 4 Helpinghelp $\\times$ Socialsocial     0.07   [0.01, 0.12]      2.32 811.70
##   p.value
## 1    .016
## 2    .024
## 3    .169
## 4    .020
##
## term     : Term
```

```
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $t$
## df       : $\\mathit{df}$
## p.value  : $p$
```

```r
p_affectNegative_helping <- ggplot(
  trial_level_affect,
  aes(x = helping, y = affectNegative_duration)
) +
  facet_grid(cols = vars(social))+
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Negative affect duration"
  )
```

```r
library(lme4)

# affectPositive_categoricalChoice ~ helping*social + (helping | subj_id) + (helping | lab_id)
m_affPos_cat_full <- glmer(
  affectPositive_categoricalChoice ~ helping * social +
    (1 | subj_id),
  data   = trial_level_affect,
  family = binomial
)

apa_m_affPos_cat_full <- apa_print(m_affPos_cat_full)



# affectNegative_categoricalChoice ~ helping*social + (helping | subj_id) + (helping | lab_id)
m_affNeg_cat_full <- glmer(
  affectNegative_categoricalChoice ~ helping * social +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

apa_m_affNeg_cat_full <- apa_print(m_affNeg_cat_full)

# surprise_categoricalChoice ~ helping*social + (helping | subj_id) + (helping | lab_id)
m_surprise_cat_final <- glmer(
  surprise_categoricalChoice ~ helping*social +
    (1 | subj_id),
  data   = trial_level_affect,
  family = binomial,
```

```
  control = glmerControl(optimizer = "bobyqa")
)
apa_m_surprise_cat_final <- apa_print(m_surprise_cat_final)
```

Emotion as a function of helping and social context

To examine how emotion displays varied as a function of helping and social context, we fit linear mixed-effects models predicting trial-level emotion measures from helping condition, social context, and their interaction.

We first tested whether average emotion ratings were more positive for the social, helping condition (`affect_average ~ helping * social + (helping | subj_id) + (1 | lab_id)`). Emotion displays were more positive (or less negative) for the helping conditions ($\hat{\beta} = 0.07$, 95% CI $[0.00, 0.15]$, $t(785.65) = 1.96$, $p = .051$). However, there was no significant effect of social vs. non-social condition ($\hat{\beta} = 0.10$, 95% CI $[-0.01, 0.20]$, $t(107.51) = 1.81$, $p = .072$), and no significant interaction between whether the condition was the social and whether it displayed a helping interaction ($\hat{\beta} = -0.09$, 95% CI $[-0.19, 0.02]$, $t(785.81) = -1.60$, $p = .111$). We conducted a follow-up analysis restricted to the first two trials per participant, which yielded a comparable pattern of results.

Next, we tested whether the duration of positive emotion displays was longer in the social, helping condition (`affectPositive_duration ~ helping * social + (1 | subj_id)`). There was no significant effect of helping ($\hat{\beta} = 0.01$, 95% CI $[-0.02, 0.04]$, $t(814.36) = 0.73$, $p = .465$), social ($\hat{\beta} = 0.02$, 95% CI $[-0.02, 0.07]$, $t(165.92) = 0.96$, $p = .337$), or their interaction ($\hat{\beta} = 0.00$, 95% CI $[-0.05, 0.04]$, $t(814.52) = -0.10$, $p = .923$) on the duration of positive emotion displays.

Finally, we examined how the duration of negative emotion displays differed across conditions (`affectNegative_duration ~ helping * social + (1 | subj_id) + (1 | lab_id)`). Displays of negative emotion had shorter duration in the helping conditions ($\hat{\beta} = -0.04$, 95% CI $[-0.08, -0.01]$, $t(811.43) = -2.26$, $p = .024$), and there was a sigificant interaction, such that this effect was weaker for the social condition ($\hat{\beta} = 0.07$, 95% CI $[0.01, 0.12]$, $t(811.70) = 2.32$, $p = .020$). As before, there was no main effect of social vs non-social condition ($\hat{\beta} = -0.04$, 95% CI $[-0.09, 0.02]$, $t(166.04) = -1.38$, $p = .169$).

```
library(patchwork)

p_affectAverage_helping +
p_affectPositive_helping +
p_affectNegative_helping
```

In addition to examining contunuous effects of condition on the avreage emotion display rating or the duration of emotion displays, we also fit models predicting the overall presence of positive, negative and surprised emotion displays by condition. There was no significant effect of helping ($\hat{\beta} = -0.06$, 95% CI $[-0.60, 0.48]$, $z = -0.21$, $p = .836$), social ($\hat{\beta} = 0.12$, 95% CI $[-0.64, 0.88]$, $z = 0.31$, $p = .758$) condition, or their interaction ($\hat{\beta} = 0.34$, 95% CI $[-0.42, 1.09]$, $z = 0.87$, $p = .383$) on the presence of positive emotion displays (`affectPositive_categoricalChoice ~ helping * social + (1 | subj_id)`). Similarly, there was no significant effect of helping ($\hat{\beta} = -0.35$, 95% CI $[-0.88, 0.18]$, $z = -1.31$, $p = .190$), social ($\hat{\beta} = -0.33$, 95% CI $[-1.11, 0.44]$, $z = -0.85$, $p = .398$), or their interaction ($\hat{\beta} = 0.61$, 95% CI $[-0.15, 1.37]$, $z = 1.57$, $p = .117$) on the presence of negative emotion displays (`affectNegative_categoricalChoice ~ helping * social + (1 | subj_id) + (1 | lab_id)`). The presence of a surprisal response (`surprise_categoricalChoice ~ helping * social + (1 | subj_id)`) was also unaffected by the helping ($\hat{\beta} = 0.19$, 95% CI $[-0.39, 0.77]$, $z = 0.64$, $p = .521$) and social ($\hat{\beta} = 0.18$, 95% CI $[-0.72, 1.09]$, $z = 0.40$, $p = .689$) conditions or their interaction ($\hat{\beta} = -0.28$, 95% CI $[-1.10, 0.55]$, $z = -0.66$, $p = .512$).

Overall, there results suggest that although the presense and duration of positive emotion displays was not shaped by the experimental conditions, participants' emotion displays were more negative in their overal magnitude and in the duration of negative displays during the hindering conditions.

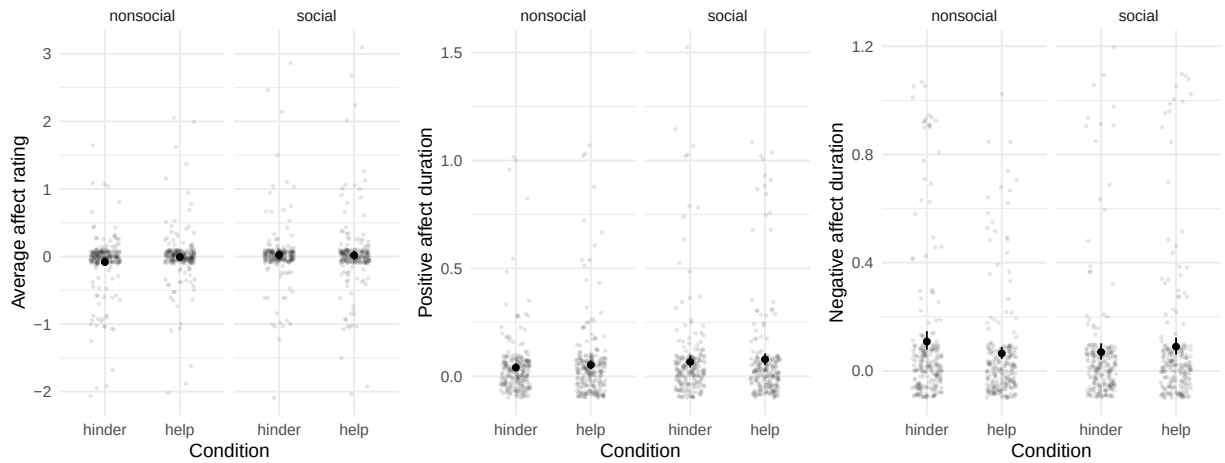


Figure 2: Caregiver affect as a function of helping condition (average affect, positive duration, negative duration)

```
# chose_helper ~affect_average*helping + (affect_average*helping | participant_id) + (affect_average*he
m_choseHelper_affectAverage_helping <- glmer(
  chose_helper ~ affect_average * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

apa_m_choseHelper_affectAverage_helping <- apa_print(m_choseHelper_affectAverage_helping)

#print("chose_helper ~affect_average*helping:")
#apa_m_choseHelper_affectAverage_helping$table


# chose_helper ~affectPositive_duration*helping + (affectPositive_duration*helping | participant_id) +
m_choseHelper_affectPositive_helping <- glmer(
  chose_helper ~ affectPositive_duration * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

apa_m_choseHelper_affectPositive_helping <- apa_print(m_choseHelper_affectPositive_helping)

#print("chose_helper ~affectPositive_duration*helping:")
#apa_m_choseHelper_affectPositive_helping$table



#chose_helper ~affectNegative_duration*helping + (affectNegative_duration*helping | participant_id) + (
m_choseHelper_affectNegative_helping <- glmer(
```

```r
  chose_helper ~ affectNegative_duration * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

apa_m_choseHelper_affectNegative_helping <- apa_print(m_choseHelper_affectNegative_helping)

#print("chose_helper ~affectNegative_duration*helping:")
#apa_m_choseHelper_affectNegative_helping$table
```

```r
library(lme4)

# chose_helper ~ positive_categoricalChoice*helping + (positive_categoricalChoice*helping | subj_id) +
m_choseHelper_posChoice_full <- glmer(
  chose_helper ~ affectPositive_categoricalChoice * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

apa_m_choseHelper_posChoice_full <- apa_print(m_choseHelper_posChoice_full)

#print("chose_helper ~ positive_categoricalChoice*helping:")
#apa_m_choseHelper_posChoice_full$table

# chose_helper ~ negative_categoricalChoice*helping + (negative_categoricalChoice*helping | subj_id) +
m_choseHelper_negChoice_full <- glmer(
  chose_helper ~ affectNegative_categoricalChoice * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)
apa_m_choseHelper_negChoice_full <- apa_print(m_choseHelper_negChoice_full)

#print("chose_helper ~ negative_categoricalChoice*helping:")
#apa_m_choseHelper_negChoice_full$table

# chose_helper ~ surprise_categoricalChoice*helping + (surprise_categoricalChoice*helping | subj_id) +
m_choseHelper_surpriseChoice_full <- glmer(
  chose_helper ~ surprise_categoricalChoice * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

apa_m_choseHelper_surpriseChoice_full <- apa_print(m_choseHelper_surpriseChoice_full)

#print("chose_helper ~ surprise_categoricalChoice*helping:")
```

```
#apa_m_choseHelper_surpriseChoice_full$table
```

Do emotion displays predict choice?

To evaluate the prediction that infants' emotional reactions during the habituation events would predict reaching choice, we fit logistic mixed-effects models with infants' choice of the helper (vs. the hinderer) toy at the end of the study as the outcome. Specifically, we tested whether infants who showed more positive, more negative, or more extreme average affect during a given trial type were more likely to choose the helper. There was no significant effect of the average emotion display rating during habituation ($\hat{\beta} = -0.33$, 95% CI $[-5.70, 5.04]$, $z = -0.12$, $p = .905$), or its interaction with the helping condition ($\hat{\beta} = 0.29$, 95% CI $[-6.68, 7.25]$, $z = 0.08$, $p = .935$) on the infant's choice of character (`chose_helper ~ affect_average * helping + (1 | subj_id) + (1 | lab_id)`). Similarly, infants' character choice (`chose_helper ~ affectPositive_duration * helping + (1 | subj_id) + (1 | lab_id)`) was not shaped by the duration of positive affect ($\hat{\beta} = -0.89$, 95% CI $[-13.22, 11.44]$, $z = -0.14$, $p = .887$) or its interaction with condition ($\hat{\beta} = 0.76$, 95% CI $[-14.31, 15.82]$, $z = 0.10$, $p = .921$). This was also no effect of negative affect duration ($\hat{\beta} = 0.01$, 95% CI $[-8.12, 8.13]$, $z = 0.00$, $p = .998$), or its interaction with condition ($\hat{\beta} = -0.28$, 95% CI $[-13.47, 12.92]$, $z = -0.04$, $p = .967$) on character choice (`chose_helper ~ affectNegative_duration * helping + (1 | subj_id) + (1 | lab_id)`).

Next, we tested whether the overall presence of positive affect, negative affect, or surprise, coded as a categorical choice, during helping versus hindering trials shaped whether infants were more likely to choose the helper (`chose_helper ~ [emotion categorical coding] * helping + (1 | subj_id) + (1 | lab_id)`). As with the continuous analyses, we found no effect of negative affect ($\hat{\beta} = 0.03$, 95% CI $[-5.30, 5.37]$, $z = 0.01$, $p = .990$) or its interaction with condition (apa_m_choseHelper_negChoice_full$full_result$affectNegative_categoricalChoice_helpinghelp), and no effect of positive affect ($\hat{\beta} = -0.06$, 95% CI $[-5.53, 5.42]$, $z = -0.02$, $p = .984$), or its interaction with condition ($\hat{\beta} = -0.01$, 95% CI $[-7.21, 7.18]$, $z = 0.00$, $p = .997$). Finally, there was no effect of the presence of a surprisal response on choice ($\hat{\beta} = 0.02$, 95% CI $[-5.73, 5.77]$, $z = 0.01$, $p = .994$), or its interaction with condition ($\hat{\beta} = -0.19$, 95% CI $[-8.00, 7.61]$, $z = -0.05$, $p = .961$)

Exploratory time-course analyses

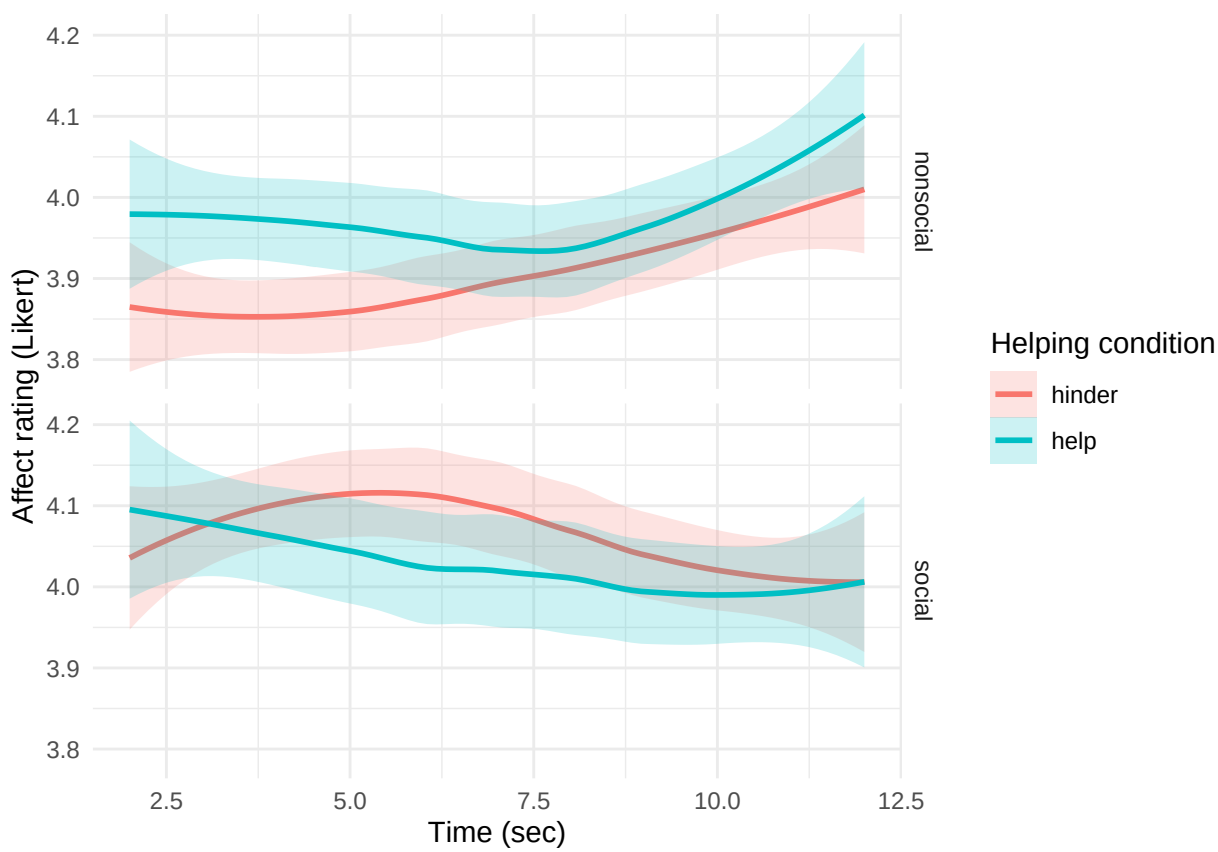
In an exploratory analysis, we examined the time course of emotion displays by transforming the event-level coding into a 1 Hz time series.

```
library(data.table)
setDT(merged_data)

# Exclude rows where start or end time is NA
merged_clean <- merged_data[
  !is.na(segment_start_sec) & !is.na(segment_end_sec)
]

# Expand into 1 Hz time series
time_series_data <- merged_clean[
  ,
  {
    # turn onset and offset into integers
    start_sec <- floor(segment_start_sec)
    end_sec   <- ceiling(segment_end_sec)
```

```r
    # generate one row per second
    seconds <- seq(start_sec, end_sec, by = 1)

    data.table(
      time_sec = seconds,
      affect_rating_segment = affect_rating_segment,
      lab_id = lab_id,
      subj_id = subj_id,
      folder = folder,
      file_name = file_name,
      segment_start_sec = segment_start_sec,
      segment_end_sec = segment_end_sec,
      duration_sec = duration_sec,
      duration_msec = duration_msec,
      happiness_full_video = happiness_full_video,
      surprise_full_video = surprise_full_video,
      trial_num = trial_num,
      age_days = age_days,
      age_group = age_group,
      condition = condition,
      cb_order = cb_order,
      helper_hinderer_choice = helper_hinderer_choice,
      push_up_identity = push_up_identity,
      push_up_order = push_up_order,
      push_up_side = push_up_side,
      meet_eligibility = meet_eligibility
    )
  },
  by = .I
]


time_series_data[, trial_num_int := as.integer(trial_num)]
time_series_data[
  ,
  is_push_up :=
    (push_up_order == "first"  & trial_num_int %% 2L == 1L) |
    (push_up_order == "second" & trial_num_int %% 2L == 0L)
]
time_series_data[
  ,
  helping := factor(
    ifelse(is.na(is_push_up), NA_character_,
           ifelse(is_push_up, "help", "hinder")),
    levels = c("hinder", "help")
  )
]

time_series_data[
  ,
  affect_rating_centered := affect_rating_segment - mean(affect_rating_segment, na.rm = TRUE),
  by = subj_id
]
```

```r
p_time_affect <- ggplot(
  time_series_data,
  aes(x = time_sec, y = affect_rating_segment, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Affect rating (Likert)",
    color = "Helping condition",
    fill  = "Helping condition"
  )

p_time_affect
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



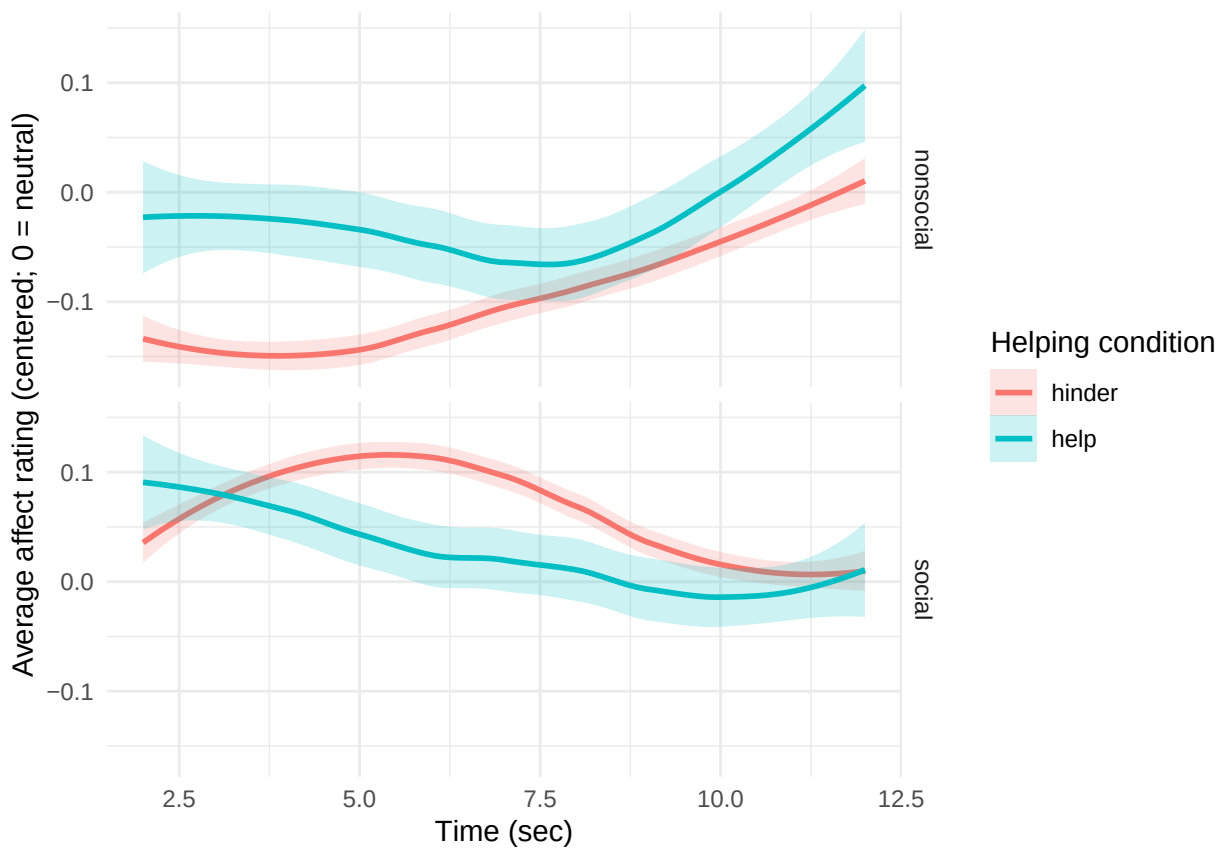
```r
library(data.table)
setDT(merged_data)

# Exclude rows where start or end time is NA
merged_clean <- merged_data[
  !is.na(segment_start_sec) & !is.na(segment_end_sec)
]

# Expand into 1 Hz time series
time_series_data <- merged_clean[
  ,
  {
    # turn onset and offset into integers
    start_sec <- floor(segment_start_sec)
    end_sec   <- ceiling(segment_end_sec)

    # generate one row per second
    seconds <- seq(start_sec, end_sec, by = 1)

    data.table(
      time_sec = seconds,
      affect_rating_segment = affect_rating_segment,

      # Neutral-centered affect (neutral=0). Original scale: 1..7 with 4 = neutral
      affect_c0 = affect_rating_segment - 4,

      # Mark each timepoint as 1/0 for proportion measures
      # (mean of these 0/1 variables within a time bin = proportion)
      nonneutral_01 = fifelse(is.na(affect_rating_segment), NA_real_,
                              as.numeric((affect_rating_segment - 4) != 0)),
      pos_01        = fifelse(is.na(affect_rating_segment), NA_real_,
                              as.numeric((affect_rating_segment - 4) > 0)),
      neg_01        = fifelse(is.na(affect_rating_segment), NA_real_,
                              as.numeric((affect_rating_segment - 4) < 0)),

      lab_id = lab_id,
      subj_id = subj_id,
      folder = folder,
      file_name = file_name,
      segment_start_sec = segment_start_sec,
      segment_end_sec = segment_end_sec,
      duration_sec = duration_sec,
      duration_msec = duration_msec,
      happiness_full_video = happiness_full_video,
      surprise_full_video = surprise_full_video,
      trial_num = trial_num,
      age_days = age_days,
      age_group = age_group,
      condition = condition,
      cb_order = cb_order,
      helper_hinderer_choice = helper_hinderer_choice,
      push_up_identity = push_up_identity,
      push_up_order = push_up_order,
```

```r
      push_up_side = push_up_side,
      meet_eligibility = meet_eligibility
    )
  },
  by = .I
]

time_series_data[, trial_num_int := as.integer(trial_num)]
time_series_data[
  ,
  is_push_up :=
    (push_up_order == "first"  & trial_num_int %% 2L == 1L) |
    (push_up_order == "second" & trial_num_int %% 2L == 0L)
]
time_series_data[
  ,
  helping := factor(
    ifelse(is.na(is_push_up), NA_character_,
           ifelse(is_push_up, "help", "hinder")),
    levels = c("hinder", "help")
  )
]

# Summarize into 1s bins to get time course measures:
# - Avg rating (centered; neutral=0)
# - Proportion not neutral (non-zero)
# - Proportion positive / negative
timecourse_1s <- time_series_data[
  ,
  .(
    avg_rating_c0   = mean(affect_c0, na.rm = TRUE),
    prop_nonneutral = mean(nonneutral_01, na.rm = TRUE),
    prop_pos        = mean(pos_01, na.rm = TRUE),
    prop_neg        = mean(neg_01, na.rm = TRUE),
    n_obs           = sum(!is.na(affect_c0))
  ),
  by = .(condition, helping, time_sec)
]

# Plot average affect rating over time (centered; 0 = neutral)
p_time_affect_avg <- ggplot(
  timecourse_1s,
  aes(x = time_sec, y = avg_rating_c0, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
```

```r
  labs(
    x = "Time (sec)",
    y = "Average affect rating (centered; 0 = neutral)",
    color = "Helping condition",
    fill  = "Helping condition"
  )

# Plot proportion non-neutral over time
p_time_affect_prop_nonneutral <- ggplot(
  timecourse_1s,
  aes(x = time_sec, y = prop_nonneutral, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Proportion non-neutral",
    color = "Helping condition",
    fill  = "Helping condition"
  )

# Plot proportion positive over time
p_time_affect_prop_pos <- ggplot(
  timecourse_1s,
  aes(x = time_sec, y = prop_pos, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Proportion positive",
    color = "Helping condition",
    fill  = "Helping condition"
  )

# Plot proportion negative over time
p_time_affect_prop_neg <- ggplot(
  timecourse_1s,
  aes(x = time_sec, y = prop_neg, color = helping)
```

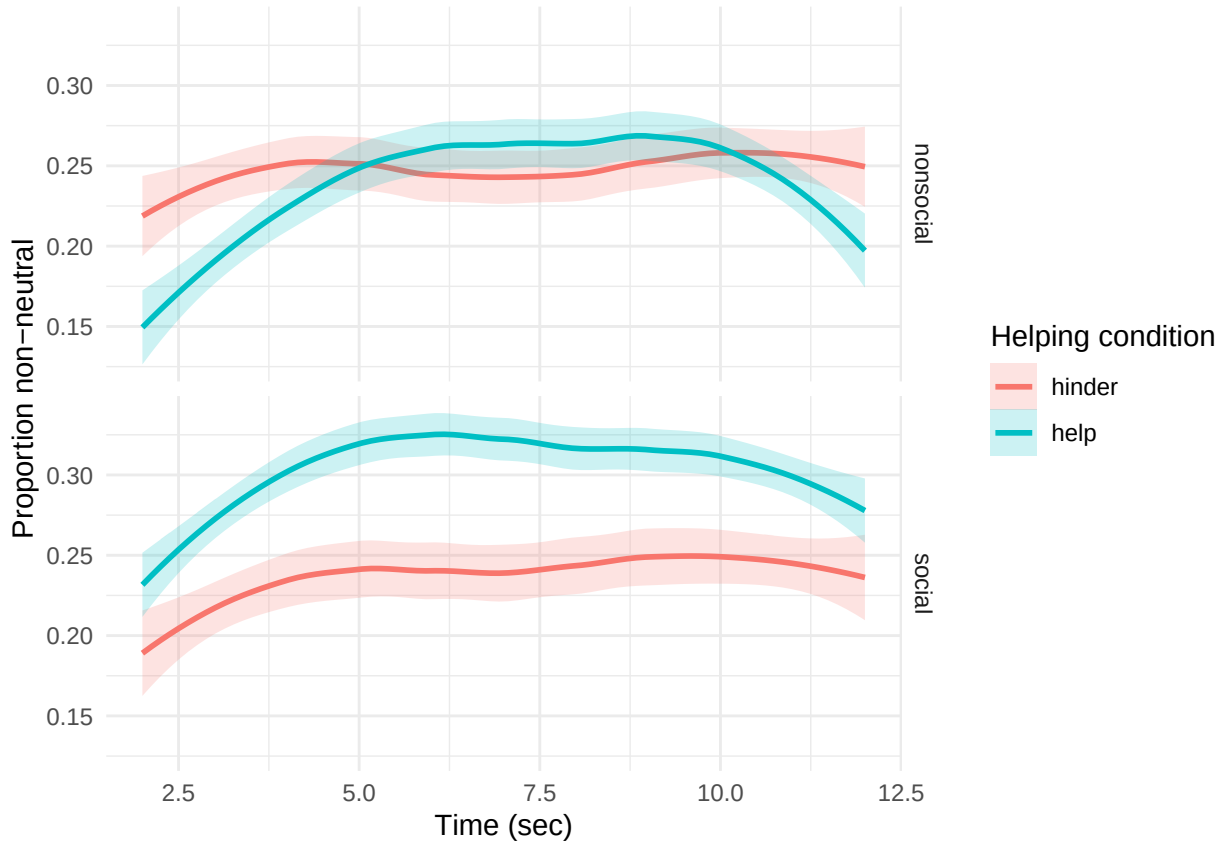
```r
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Proportion negative",
    color = "Helping condition",
    fill  = "Helping condition"
  )

# Print plots
p_time_affect_avg
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



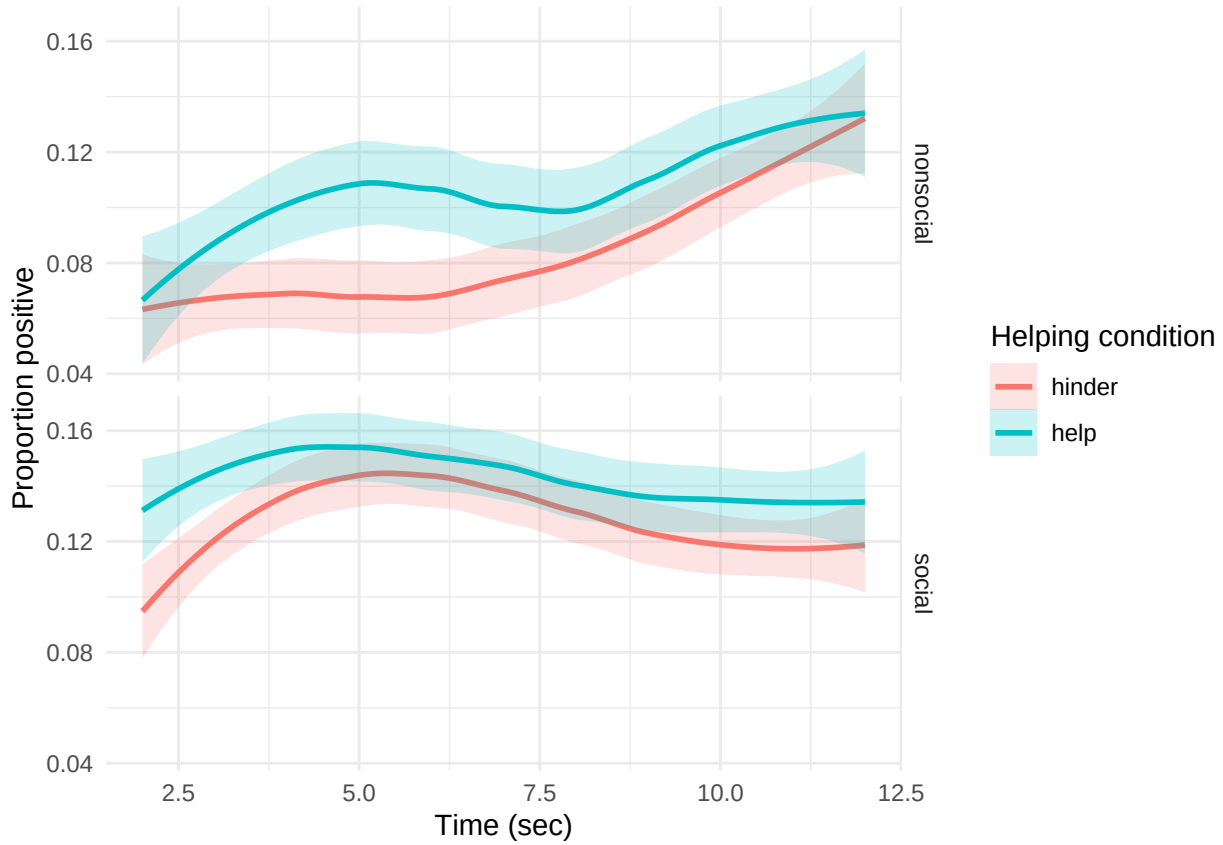

```r
p_time_affect_prop_nonneutral
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



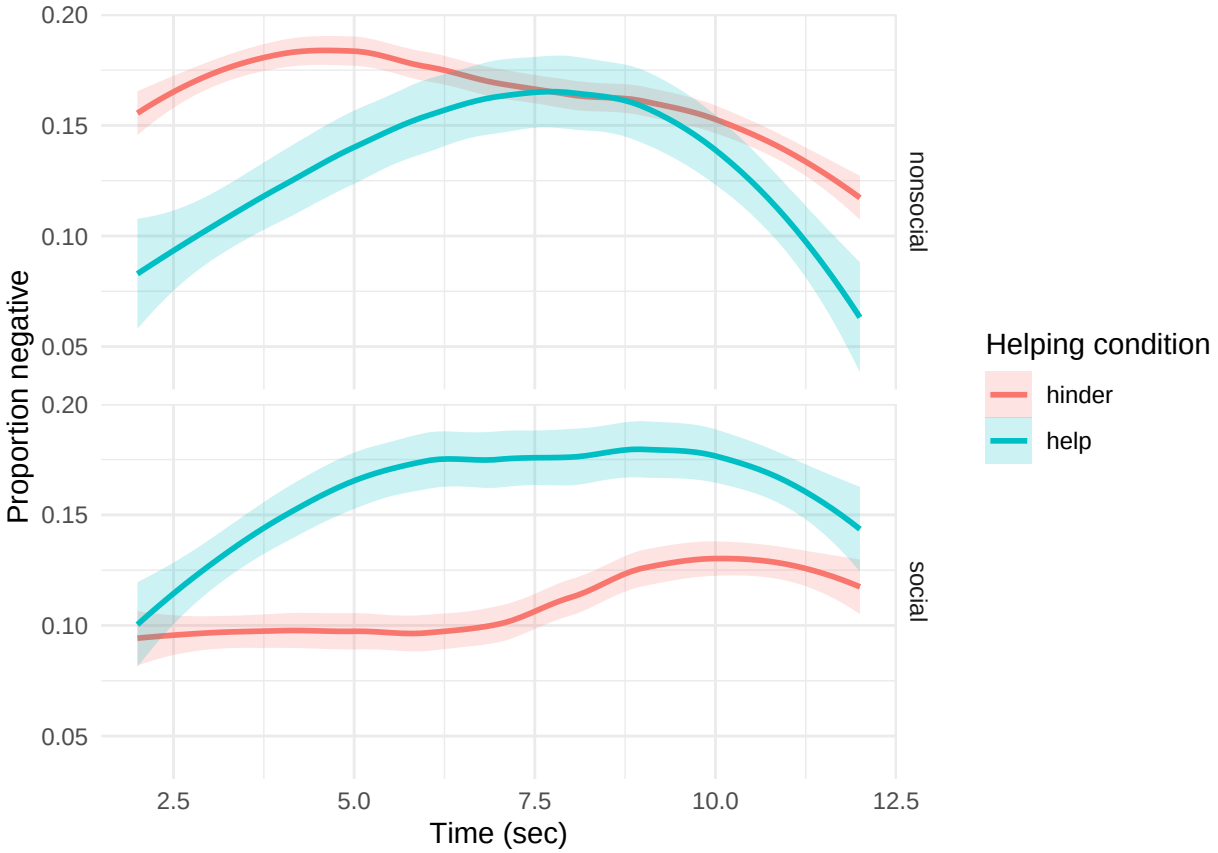

```
p_time_affect_prop_pos
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
p_time_affect_prop_neg
```

```
## `geom_smooth()` using formula = 'y ~ x'
```


0.20
0.15
0.10
0.05
Proportion negative
nonsocial
social
2.5
5.0
7.5
10.0
12.5
Time (sec)
Helping condition
hinder
help